

Version comparison: EXHALE v1.0 vs current (He/H + metal diffusion)

on HD 209458b and WASP-121b

1. Purpose and setup

Quantify what the new diffusive separation changes in EXHALE's predictions, by running the *same* planet on two code versions, for *two contrasting planets*:

- **v1.0** (ATES/EXHALE_v1.0/): clean GitHub main + the temperature-dependent Penning rate. *No* diffusive separation.
- **current** (ATES/EXHALE/): v1.0 physics + He/H and per-element metal diffusion (He_diffusion + He_metal_diffusion, ambipolar on).

Both share the temperature-dependent Penning rate, so the only difference is the diffusion. Setup: He 2^3S on, trace metals from metals.inp (HD 209458b: C/N/O; WASP-121b: C/N/O/Mg/Ca/Na/Fe), identical input.inp per planet, 15000 relaxation steps. Runs/figures/script: docs/version_compare/.

2. Headline numbers

HD 209458b (hot Jupiter, $\log_{10} \dot{M} \approx 9.3$ — gentler escape)

quantity	v1.0 (no diff)	current (diff)	change
He/H at $3 R_p$	0.083 (flat)	0.014	−83%
peak $n(\text{He } 2^3\text{S})$ [cm^{-3}]	177	102	−42%
He I 10830 line-center absorption	70.5%	27.2%	−2.6×
$\log_{10} \dot{M}$ [g s^{-1}]	9.31	9.04	−0.27 dex

WASP-121b (ultra-hot Jupiter, $\log_{10} \dot{M} \approx 13.3$ — furious escape)

quantity	v1.0 (no diff)	current (diff)	change
He/H (min, $1.05\text{--}1.5 R_p$)	0.083	0.053	−36% (weaker)
peak $n(\text{He } 2^3\text{S})$ [cm^{-3}]	3517	2383	−32%
He I 10830 line-center absorption	53.7%	53.6%	~unchanged
metal/H (elem/base)	1.0	tracks He ($\sim 0.6\text{--}1.0$; no extra fractionation)	none
$\log_{10} \dot{M}$ [g s^{-1}]	13.34	13.32	unchanged

Two regimes. On the gentler HD 209458b the escape is slow enough that *helium itself* diffusively separates, cutting the He I 10830 line by $\sim 2.6\times$, and the heavier C/N/O deplete more than He aloft ($\text{He} > \text{C} > \text{N} > \text{O}$ at $3 R_p$). On the

furious WASP-121b (~ 4 dex stronger wind) advection dominates settling for *every* species — helium *and* the metals are dragged out essentially unfractionated (10830 unchanged; metal/H tracks He/H). This is the (settling speed)/(wind speed) scaling of Koskinen et al. (2013) / Xing et al. (2023): separation is suppressed by fast escape and, where it operates at all, is stronger for heavier species.

Correction note: an earlier version showed WASP-121b metals collapsing to zero by $\sim 1.25 R_p$ (a “metal homopause”). That was an artifact of two bugs in the metal rescale — an $r_X \leq 1$ clamp that made depletion a one-way ratchet, and a skip of exhausted cells that made holes permanent — which locked in transient early-relaxation settling before the wind developed. With the fix, metals recover with the wind and track He, and the v1-vs-v2 temperature structures agree; the earlier “ ~ 2500 K hotter” claim is also retracted.

3. HD 209458b

HD 209458b: EXHALE v1.0 (no diffusion) vs current (He/H + metal diffusion)

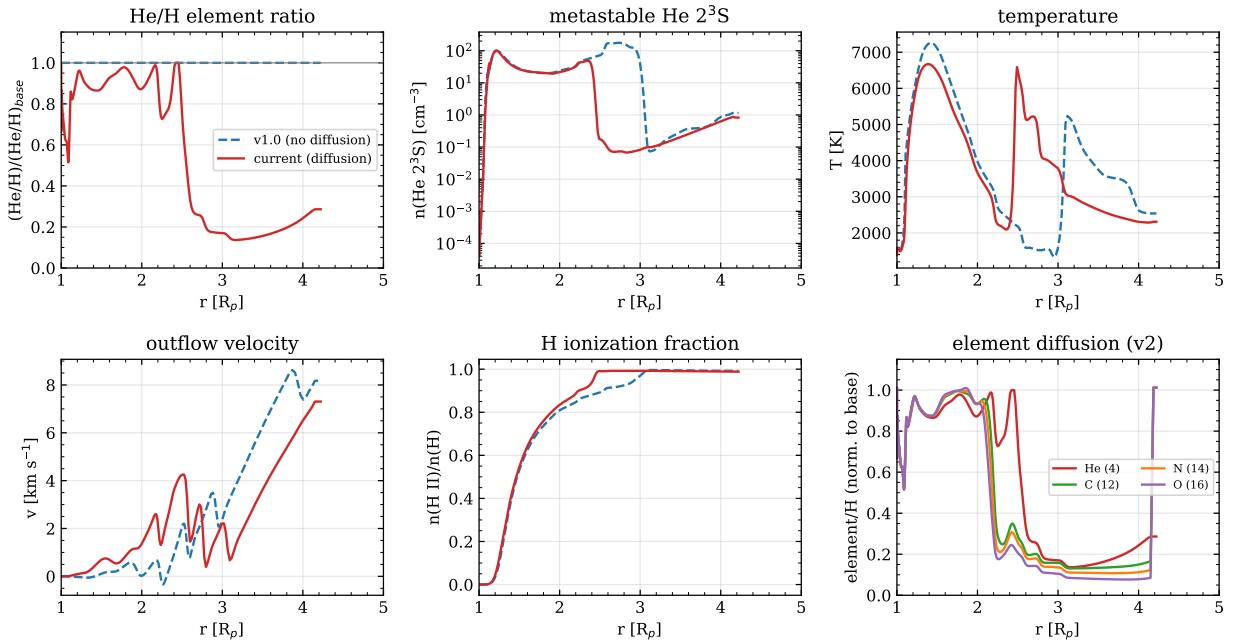


Figure 1: HD 209458b, v1.0 (blue dashed) vs current (red). He/H falls to ~ 0.15 – $0.3\times$ aloft; the $\text{He } 2^3\text{S}$ reservoir contracts. Bottom-right: C/N/O deplete more than He aloft (mass ordering $\text{He} > \text{C} > \text{N} > \text{O}$ above $\sim 2.5 R_p$).

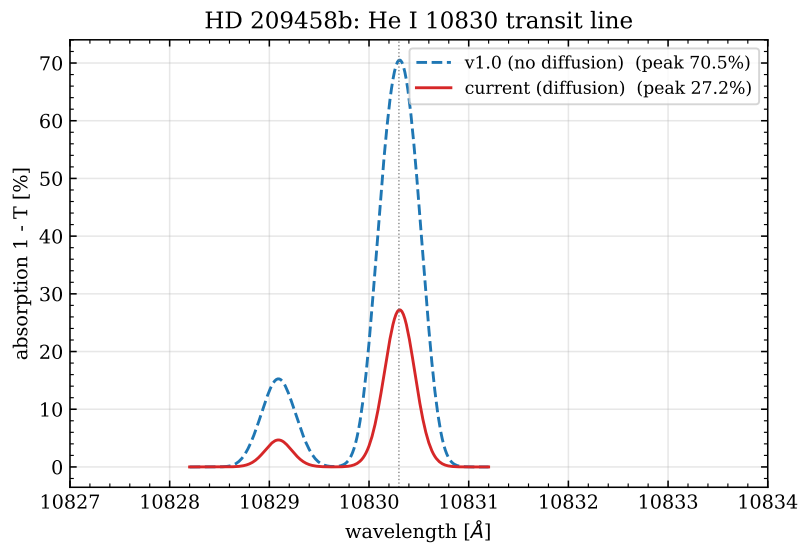


Figure 2: He I 10830 line: diffusion lowers the line-center absorption 70.5% $\rightarrow$ 27.2% and narrows the profile.

4. WASP-121b

WASP-121b: EXHALE v1.0 (no diffusion) vs current (He/H + metal diffusion)

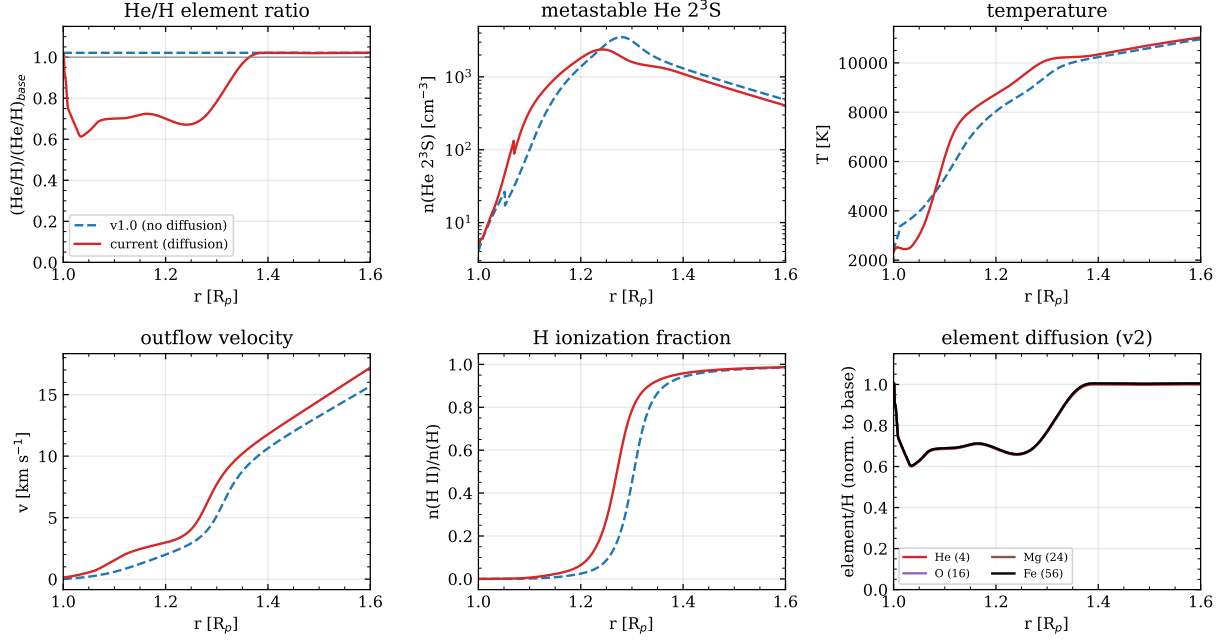


Figure 3: WASP-121b (compact domain, escape radius $1.5 R_p$). He/H separates only modestly (~ 0.6 – $0.75\times$, returning to ~ 1 above $1.4 R_p$) because the ~ 4 -dex-stronger wind drags helium out. The metals (O, Mg, Fe) overlap He almost exactly — even Fe (mass 56) is advection-dominated here, so no additional fractionation develops. The v1/v2 temperature and velocity structures are nearly identical.

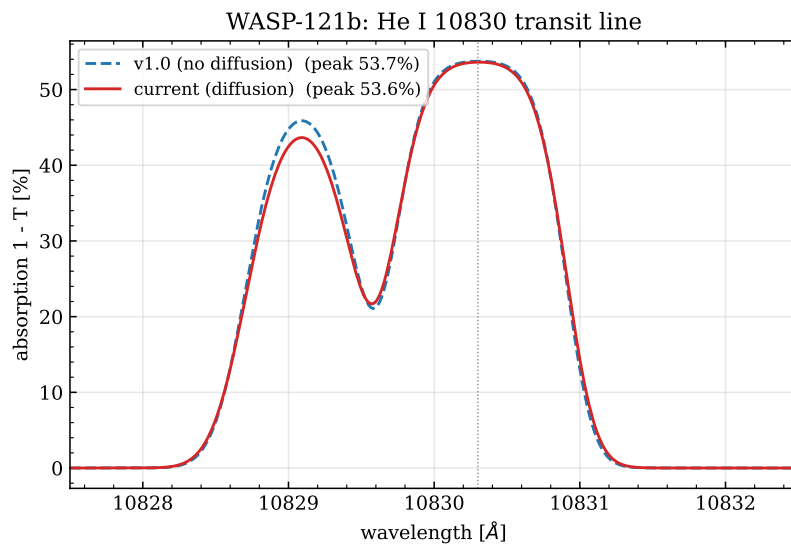


Figure 4: He I 10830 line: the core is essentially unchanged (53.7%→53.6%) since helium is not strongly separated in the line-forming region.

5. Interpretation and caveats

Interpretation.

- **He/H.** On HD 209458b settling competes with the slow wind, so He/H declines with altitude and 10830 weakens; on WASP-121b the wind is ~ 4 dex stronger, so helium is dragged out nearly unfractionated.
- **He 2³S / 10830.** The metastable tracks He abundance — weakened where He is depleted (HD 209458b), preserved where He stays mixed (WASP-121b).
- **Metals.** Each trace metal diffuses independently with its own mass, binary D , and ambipolar-corrected settling. Where diffusion competes with the wind (HD 209458b aloft) heavier elements deplete faster (He > C > N > O); where advection dominates (WASP-121b) even Fe stays locked to He. Relevant for metal-line transmission (Mg II/Ca II/Na I in TPM.py).

Caveats.

- **Absolute values uncalibrated.** The 27–70% He 10830 depths far exceed observations (~ 1 –2%); heating efficiency / XUV / He 2³S microphysics were not tuned. The *relative* change and the two-planet *contrast* are the robust results.
- **Scheme / convergence sensitivity.** Step-capped (not Newton-finished), so $\dot{M}$ is indicative; the He-separation magnitude is somewhat sensitive to the settling discretization (a Peclet-hybrid central/upwind scheme: central where well-resolved, upwind for the stiff heavy-metal settling).
- **Phase-1/2 numerics.** Mild non-monotonic structure from the He/H cap and the not-fully-developed wind; see `design_hehe_diffusion.md` §7d–§7e. Diffusion flags **default OFF**.